Coding & Solution

Date	30 June 2026
Team ID	SWTID-2026-4993
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Yaddala-Geethanjali/OptiCrop-Project
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	Flask, Scikit-learn
Key Features	Crop prediction, ML model integration, responsive UI,

Implemented	confidence score
Pending / Incomplete Features	Cloud deployment, database integration, live weather API
Setup / Run Instructions	Install dependencies → Run python app.py → Open localhost in browser

Code Quality Checklist

S.No	**Criteria**	**Status (Yes / No)**
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments

- Developed an AI-powered crop recommendation system using Machine Learning.
- Implemented the frontend using HTML/CSS with mobile responsiveness.
- Flask backend handles user input, preprocessing, prediction, and result generation.
- The model predicts the best crop based on:
    - Nitrogen (N)
    - Phosphorus (P)
    - Potassium (K)
    - Temperature
    - Humidity
    - pH
    - Rainfall